Higher National Diploma in Engineering (Electrical)
Fourth Year, First Semester Examination – 2020
EE 3206-Robotics and Automation

Instructions for Candidates:
Answer **any Five (05)** questions.
All question carry equal marks

No. of questions : 06
No. of pages : 03
Time : **03 hours**

Q1. a. Write short notes on the oxidation and packaging of the silicon wafer manufacturing process?

(5 marks)

b. What do you meant by crystal growing? Explain Czochralski (CZ) method or Float zone (FZ) method of crystal growing?

(8 marks)

c. Distinguish between intrinsic semiconductor and extrinsic semiconductor?

(3 marks)

d. State and explain the silicon wafer preparation stages?

(4 marks)

Q2 a. Define a robot and Name the most commonly used robot configuration system?

(4 marks)

b. With help of a sketch describe the pitch, yaw and roll motion of a robot wrist?

(4 marks)

c. Briefly explain point-to-point control robot and continuous path control robot based on the speed of motion?

(6 marks)

d. Write down Asimov's law of robotics and mention any three advantages and disadvantages of robots?

(6 marks)

Q3. a. Define manipulator kinematics

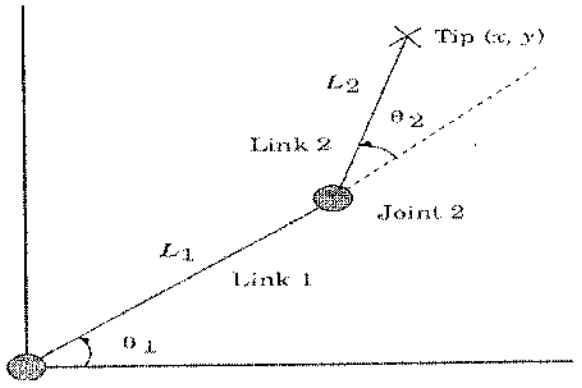
(2 marks)

b. A planner mechanisms has 9 link and 10 rotary join. Find the number of degree of freedom using Gurublers equation?

(4 marks)

c. Below figure shows the two degree of freedom manipulator if we know the link length L1 and L2 derive the expression to find the joint angle q= (Θ_1, O_2) of the inverse kinematics to place the arm at a position p= (x, y)?

(8 marks)



d. (i) State the benefit of industrial robot?

(3 marks)

(ii) What is meant by accuracy of robot?

(3 marks)

Q4. a. Sketch and Describe the any two of the commonly used robot cell layout in the industry with example?

(6 marks)

b. With neat sketch explain the working principle of the

(i) Internal gripper

(ii) External Gripper

(2*3=6 marks)

c. What are the general characteristics that make potential robot application technically practical and economically feasible?

(3 marks)

d. What are major types of joint used in robots and explain any of the two joints?

(5 marks)

Q5. a. i. Explain Manual lead through programming with example.

(4 marks)

ii. State the limitation of manual lead through programming method.

(3 marks)

b. What is meant by Industrial Automation and what are its types?

(2+3 = 5 marks)

c. Define control system of a robot?

(4 marks)

d. A motor gear has 24 teeth and revolves at 100 rev/min. The driven gear has 8 teeth. Calculate its rotational speed?

(4 marks)

Q6. a. What are the differences between sensor and transducer?

(3 marks)

b. Write the short notes on following sensor and also mention how they are used in robots?

i. Proximity sensor

ii. Range sensor

iii. Sound sensor

(3 * 3 = 9 marks)

c. Write down four (04) common imaging devices used for robot vision systems.

(4 marks)

d. i. Write down two (02) feedback device used in robotics?

(2 marks)

ii. What are the type of encoder?

(2 marks)